

MERIN JOHNY

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PROFESSIONAL SUMMARY

AI Engineer I with experience in AI-related technology projects, specializing in machine learning and deep learning technology (CNNs), model training, and deployment across the SDLC — developing AI algorithms and models, RAG-based chatbots, and generative models with Python, TensorFlow, and PyTorch, integrating AI solutions into existing systems within agile environments, and contributing to continuous improvement of AI development processes.

EDUCATION

Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence) 2022–2026
Adi Shankara Institute of Engineering and Technology, Kalady — CGPA: 9.27

Higher Secondary 2022
Mar Augustine's HSS, Thuravoor — 98.5%

PROFESSIONAL EXPERIENCE

AI-QC Tool Development – GP3 Cloud Innovations Jan 2026 – Mar 2026

- Assisted in developing AI algorithms and models to automate manual before/after comparison of fiber and coax network maps, reducing analysis time from several hours to under 30 minutes per map through an AI-driven pipeline.
- Tiled large maps, annotated symbols, and trained a YOLO model for symbol recognition; integrated EasyOCR for text extraction from new maps.
- Trained an AI model to detect map changes, generate callouts, and perform calculations, supporting implementation and testing of AI solutions across the SDLC.
- Built and deployed the pipeline using FastAPI and Docker, integrating the AI solution into client workflows.
- Held weekly client discussions and documentation reviews, iterating on feedback in a fast-paced environment.

AI & Web App Intern – Jorim Technology Solutions Pvt Ltd, Chennai May 2025 – Jul 2025

- Built Student Helper Hub, a Flask-based AI app with REST APIs, NLP features, and structured SQL data models, contributing to integrating AI solutions into existing systems using modular architecture.
- Automated testing and model performance tuning, identified bottlenecks, and collaborated in an agile development environment using Agile/Scrum to deliver a production-ready solution.

PROJECTS

VirtuTeach – AI-Based Personalized Learning Platform

- Designed and developed AI algorithms and models for an AI-native full-stack solution integrating LLMs, LangChain-based RAG pipelines, and Agentic AI workflows using Gemini API and FAISS vector database for context-aware, autonomous multi-step reasoning.
- Evaluated and fine-tuned open-weight LLMs using LoRA; built scalable backend APIs with Python and Flask, integrating document processing, vector search, and iterative evaluation pipelines.

DriverGuard – Real-Time Driver Monitoring System

- Built a real-time AI-powered Driver Monitoring System using Python, OpenCV, and YOLO for drowsiness and distraction detection; improved model accuracy through PyTorch-based model training, evaluation, and optimization.
- Conducted root-cause analysis, validated software functionality, and executed structured enhancement cycles with strong attention to detail to ensure production-ready reliability; managed code using Git and GitHub with versioning and pull requests.

VocalPen

- Built an end-to-end AI-powered application integrating image preprocessing, OCR, audio generation, REST API integration, and scalable data processing workflows.
- Ensured production-ready reliability through systematic debugging, automated testing, and technical documentation, contributing to continuous improvement of AI development processes.

SKILLS

AI: Agentic AI, LangChain, LlamaIndex, OpenAI API, Gemini API, Claude, FAISS, Vector Databases

Programming: Python, TensorFlow, PyTorch, SQL, JavaScript, OOP, Data Structures & Algorithms

Core Foundations: Mathematics

Backend & APIs: FastAPI, Flask, REST APIs, JSON

Machine Learning & Deep Learning Technology: Machine Learning, Deep Learning, CNNs, NLP, Transformers, Generative AI, LLMs, RAG, Model Fine-tuning, LoRA, Model Training & Evaluation, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, OpenCV, YOLO

Databases & Cloud: MySQL, PostgreSQL, MongoDB, Redis, Git, GitHub, CI/CD, Docker, AWS

Software Engineering: Agile/Scrum, Agile Development Environment, SDLC, Automated Testing, Debugging, Deployment, Technical Documentation

Soft Skills: Problem Solving, Problem-Solving Capability, Attention to Detail, Analytical Thinking, Quick Learning, Communication, Collaboration, Fast-Paced and Dynamic Environment

CERTIFICATIONS & ACHIEVEMENTS

- Microsoft – Foundations of AI and ML (Jul 2025)
- AWS – Cloud Technical Essentials (Jun 2025)
- IIT Ropar NPTEL – Deep Learning (Oct 2025)
- IBM – Data Analytics Essentials (Mar 2025)
- Presented "DriverGuard" at ICIMRBE 2025
- Prize Winner – Innovative Project (VocalPen)
- Best Outgoing Student – Department of AI & DS,(2026)
- TinkerHack Women's Hackathon – Participant
- Newsletter Head & Placement Cell Coordinator
- Coordinator – Project Expo, Prayag Technical Fest, Brahma National Fest.